

Logic and Reasoning Mastery Workbook 2

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I have neither given nor received unauthorized assistance.

1. (10pts) Before anything else, annotate examples 7 and 8 from page 30. You MUST use this EXACT format when showing logical equivalences (and proofs) to get credit for your work in this course.
(Insert image here)

2. (25pts) We will prove the first 5 logical equivalences in Chart 7 page 28

- USE only RESULTS FROM Chart 2 and Chart 6 to justify steps.

When making logical chains of equivalences use the REQUIRED format.

- Prove these in this order and you may use the results as you go.
- This first one is what Chris calls RBI and it is essential because it converts a conditional into a disjunction. Prove this with just a truth table.

$$p \rightarrow q \equiv \neg p \vee q$$

Let's structure the truth table as follows:

p	q	$p \rightarrow q$	$\neg p$	$\neg p \vee q$	Explanation
T	T	T	F	T	For this set of truth values,
T	F	F	F	F	In the case that we have p true and q false, then $p \rightarrow q$ is false, since both negation of p and q are false, so $\neg p \vee q$ is also false. Since both statements are false then they are equivalent
F	T	T	T	T	In the case that we have p false and q true, then $p \rightarrow q$ is true, and since p is false, $\neg p$ is true, so $\neg p \vee q$ is also true. Since both statements are true then they are equivalent
F	F	T	T	T	In the case that we have p false and q false, then $p \rightarrow q$ is true, and since p is false, $\neg p$ is true, so $\neg p \vee q$ is also true. Since both statements are true then they are equivalent

Since in all cases the truth values of $p \rightarrow q$ and $\neg p \vee q$ are the same, we can conclude that they are logically equivalent. (correct but come back to this to revise explanation)

Carefully repeat the proof of the following using the method without a truth table from the video. Take note of what tricks Chris uses - as we will need these as we go. (Challenge - can you prove it another way?)

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

(watch the lecture)

- Third one (hint: RBI works both ways.)

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

(watch the lecture)

- Third one (hint: RBI works both ways.)

$$p \vee q \equiv \neg(p \rightarrow q)$$

- Fourth one, (think about which rule “flips” conjunctions and disjunctions)

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

- Finally, for the fifth one, see if you can do it on your own.

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

3. (10pts) We can use Table 6 p.27 to prove new “rules.” Let’s prove a logic version of the algebraic version FOIL. You will need to use a compound substitution with an equivalence from Table 6 twice. (optional/recommended create a truth table as well).

$$(a \wedge b) \vee (c \wedge d) \equiv (a \vee c) \wedge (b \vee c) \wedge (a \vee d) \wedge (b \vee d)$$

To prove this statement, we will use the rules in Table 6. We have

$$\begin{aligned} (a \wedge b) \vee (c \wedge d) &\equiv (c \vee a) \wedge (c \vee b) \wedge (d \vee a) \wedge (d \vee b) && \text{Original Statement, rearranged...} \\ &\equiv ((c \vee a) \wedge (c \vee b)) \wedge ((d \vee a) \wedge (d \vee b)) && \text{First, using laws of associativity...} \\ &\equiv (c \vee (a \wedge b)) \wedge (d \vee (a \wedge b)) && \text{Using the law of distribution, we have...} \end{aligned}$$

Now that we have this expression, if we let $S \equiv (a \wedge b)$, then if we substitute S , we can simplify to:

$$\begin{aligned} (c \vee (a \wedge b)) \wedge (d \vee (a \wedge b)) &\equiv (c \vee S) \wedge (d \vee S) \\ &\equiv S \vee (c \wedge d) && \text{Using distributive laws again...} \\ &\equiv (a \wedge b) \vee (c \wedge d) && \text{Resubstituting the definition of } S \end{aligned}$$

Thus, we have proved $(a \wedge b) \vee (c \wedge d)$ and $(a \vee c) \wedge (b \vee c) \wedge (a \vee d) \wedge (b \vee d)$ are logically equivalent. □

4. (10 pts) Rosen, page 53. Do #8 as warm-up on your own. Do #9 below. Add notes and insight for full credit.

Problem 9 from Rosen page 53:

Let $P(x)$ be the statement "x can speak Russian" and let $Q(x)$ be the statement "x knows the computer language C++". Express each of these statements in terms of $P(x)$, $Q(x)$, quantifiers, and logical connectives. The domain of quantifiers consists of all students at your school.

- (a) There is a student at your school who can speak Russian and who knows C++.

Given that $P(x)$ is "x can speak Russian" and $Q(x)$ is "x knows the computer language C++", we can express the above statement as follows:

$$\exists x(P(x) \wedge Q(x))$$

Since we are expressing the statement of the existence of at least one student who can speak Russian AND knows C++, we use the existential quantifier $\exists$ along with the logical conjunction $\wedge$ to combine the two predicates $P(x)$ and $Q(x)$.

- (b) There is a student at your school who can speak Russian but who doesn't know C++.

Given that $P(x)$ is "x can speak Russian" and $Q(x)$ is "x knows the computer language C++", we can express the above statement as follows:

$$\exists x(P(x) \wedge \neg Q(x))$$

Since we are expressing the statement of the existence of at least one student who can speak Russian AND does NOT know C++, we use the existential quantifier $\exists$ along with the logical conjunction $\wedge$ to combine the predicate $P(x)$ and the negation of $Q(x)$, denoted as $\neg Q(x)$.

- (c) Every student at your school either can speak Russian or knows C++.

Given that $P(x)$ is "x can speak Russian" and $Q(x)$ is "x knows the computer language C++", we can express the above statement as follows:

$$\forall x(P(x) \vee Q(x))$$

Since we are expressing the statement that every student at your school either can speak Russian OR knows C++, we use the universal quantifier $\forall$ along with the logical disjunction $\vee$ to combine the two predicates $P(x)$ and $Q(x)$.

- (d) No student at your school can speak Russian or knows C++.

Given that $P(x)$ is "x can speak Russian" and $Q(x)$ is "x knows the computer language C++", we can express the above statement as follows:

$$\forall x(\neg P(x) \wedge \neg Q(x))$$

Since we are expressing the statement that no student at your school can speak Russian AND does NOT know C++, we use the universal quantifier $\forall$ along with the logical conjunction $\wedge$ to combine the negation of $P(x)$, denoted as $\neg P(x)$, and the negation of $Q(x)$, denoted as $\neg Q(x)$.

5. (10pts) #33 section 1.5, page 67 (add insights and comments for full credit). Problem 33 from Rosen page 67:

Rewrite each of these statements so that negations appear only within predicates (that is, so that no negation is outside a quantifier or an expression involving logical connectives).

Problems from #33:

(a) $\neg \forall x \forall y P(x, y)$

To simplify this expression, we establish what this statement without negation translates to in English. The statement $\forall x \forall y P(x, y)$ translates to "Statement $P(x, y)$ is true for all x and for all y ." From our textbook, we know that by definition this is also a universal quantification statement. So, when we add negation to the statement $\forall x \forall y P(x, y)$, we are performing negation on a universal quantifier and therefore:

$$\neg \forall x \forall y P(x, y) \equiv \exists x \exists y \neg P(x, y)$$

(b) $\neg \forall y \exists x P(x, y)$

Again, when we express this without negation, the statement $\forall y \exists x P(x, y)$ translates to "For every y , there exists an x such that $P(x, y)$ is true." Therefore, the negation $\neg \forall y \exists x P(x, y)$ translates to "It is not true that for every y , there exists an x such that $P(x, y)$ is true." This means there exists at least one value of y for which there is no corresponding value of x that makes $P(x, y)$ true. We can express this using the existential quantifier as follows:

$$\exists y \forall x \neg P(x, y)$$

(c) $\neg \forall y \forall x (P(x, y) \vee Q(x, y))$

For this negation, given that we have the relation $P(x, y) \vee Q(x, y)$ as nested quantifiers, to simplify our expression we will use DeMorgan's laws. DeMorgan's laws state that:

$$\neg(p \wedge q) \equiv \neg p \vee \neg q \quad (1)$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q \quad (2)$$

In this case, we can use the second DeMorgan law (5) to simplify our expression. With this and from what we proved previously in a) and b) on the negation of universal quantifiers we can simplify the expression to:

$$\exists y \exists x (\neg P(x, y) \wedge \neg Q(x, y));$$

and therefore

$$\neg \forall y \forall x (P(x, y) \vee Q(x, y)) \equiv \exists y \exists x (\neg P(x, y) \wedge \neg Q(x, y)).$$

(d) $\neg (\exists x \exists y P(x, y) \wedge \forall x \forall y Q(x, y))$

For this statement, we can rewrite it using a combination of DeMorgan's laws as well as what we proved to rewrite statements a) and b). To show this, first let $R(x, y) \equiv \exists x \exists y P(x, y)$ and $S(x, y) \equiv \forall x \forall y Q(x, y)$. When applying these substitutions and DeMorgan's law (5), we have

$$\begin{aligned} \neg (\exists x \exists y P(x, y) \wedge \forall x \forall y Q(x, y)) &\equiv \neg (R(x, y) \wedge S(x, y)) \\ &\equiv \neg R(x, y) \vee \neg S(x, y) \text{ by DeMorgan's law} \\ &\equiv \neg (\exists x \exists y P(x, y)) \vee \neg (\forall x \forall y Q(x, y)) \text{ resubstituting } R(x, y) \text{ and } S(x, y) \text{'s values} \\ &\equiv \forall x \forall y \neg P(x, y) \vee \exists x \exists y \neg Q(x, y) \text{ after applying negation to terms} \end{aligned}$$

Therefore

$$\neg (\exists x \exists y P(x, y) \wedge \forall x \forall y Q(x, y)) \equiv \forall x \forall y \neg P(x, y) \vee \exists x \exists y \neg Q(x, y)$$

$$(e) \neg \forall x (\exists x \forall z P(x, y, z) \wedge \exists z \forall y P(x, y, z))$$

For (e), we can simplify using the same tools that we used for (d). So, let $A(x, y, z) \equiv \exists x \forall z P(x, y, z)$ and $B(x, y, z) \equiv \forall x \forall y Q(x, y)$, then we have

$$\begin{aligned} \neg \forall x (\exists x \forall z P(x, y, z) \wedge \exists z \forall y P(x, y, z)) &\equiv \neg \forall x (A(x, y, z) \wedge B(x, y, z)) && \text{Applying substitution...} \\ &\equiv \exists x \neg (A(x, y, z) \wedge B(x, y, z)) && \text{first negating for all x...} \\ &\equiv \exists x (\neg A(x, y, z) \vee \neg B(x, y, z)) && \text{applying DeMorgan's laws...} \\ &\equiv \exists x (\forall x \exists z \neg P(x, y, z) \vee \forall z \exists y \neg P(x, y, z)) && \text{resubstituting...} \end{aligned}$$

Therefore,

$$\neg \forall x (\exists x \forall z P(x, y, z) \wedge \exists z \forall y P(x, y, z)) \equiv \exists x (\forall x \exists z \neg P(x, y, z) \vee \forall z \exists y \neg P(x, y, z))$$

6. (35 pts) Now let's go back to section 1.1. Read the examples on page 11 and 12 to see an important computer science application. Do each of #43 problems and check your answers in the back of the book. Add notes and insights.

43. Find the bitwise OR, bitwise AND, and bitwise XOR of each of these pairs of bit strings.

Note: before finding the answers to the bitwise OR, AND, and XOR of each pair, we will be using the following truth table frequently:

x	y	$x \wedge y$	$x \vee y$	$x \oplus y$
0	0	0	0	0
0	1	0	1	1
1	0	0	1	1
1	1	1	1	0

As stated in the book, truth values can also be translated into binary zeros and ones, and the definitions of each bitwise operations are (if x and y are binary strings):

$$\text{AND} := x \wedge y$$

$$\text{OR} := x \vee y$$

$$\text{XOR} := x \oplus y$$

(a) 101 1110, 010 0001

AND	OR	XOR
1 0 1 1 1 1 0	1 0 1 1 1 1 0	1 0 1 1 1 1 0
0 1 0 0 0 0 1	0 1 0 0 0 0 1	0 1 0 0 0 0 1
0 0 0 0 0 0 0	1 1 1 1 1 1 1	1 1 1 1 1 1 1

(b) 1111 0000, 1010 1010

AND	OR	XOR
1 1 1 1 0 0 0 0	1 1 1 1 0 0 0 0	1 1 1 1 0 0 0 0
1 0 1 0 1 0 1 0	1 0 1 0 1 0 1 0	1 0 1 0 1 0 1 0
1 0 1 0 0 0 0 0	1 1 1 1 1 0 1 0	0 1 0 1 1 0 1 0

(c) 00 0111 0001, 10 0100 1000

AND	OR	XOR
0 0 0 1 1 1 0 0 0 1	0 0 0 1 1 1 0 0 0 1	0 0 0 1 1 1 0 0 0 1
1 0 0 1 0 0 1 0 0 0	1 0 0 1 0 0 1 0 0 0	1 0 0 1 0 0 1 0 0 0
0 0 0 1 0 0 0 0 0 0	1 0 0 1 1 1 1 0 0 1	1 0 0 0 1 1 1 0 0 1

(d) 11 1111 1111, 00 0000 0000

AND	OR	XOR
1 1 1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1 1 1
0 0 0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0 0 0	0 0 0 0 0 0 0 0 0 0
0 0 0 0 0 0 0 0 0 0	1 1 1 1 1 1 1 1 1 1	1 1 1 1 1 1 1 1 1 1